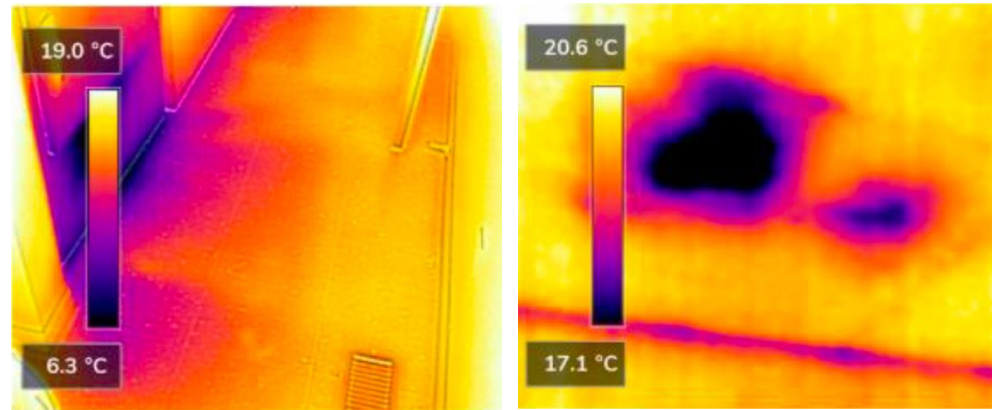


Sea surface temperature (SST) From infrared data

Andrea Scott, Associate Professor
Mechanical and Mechatronics Engineering
University of Waterloo

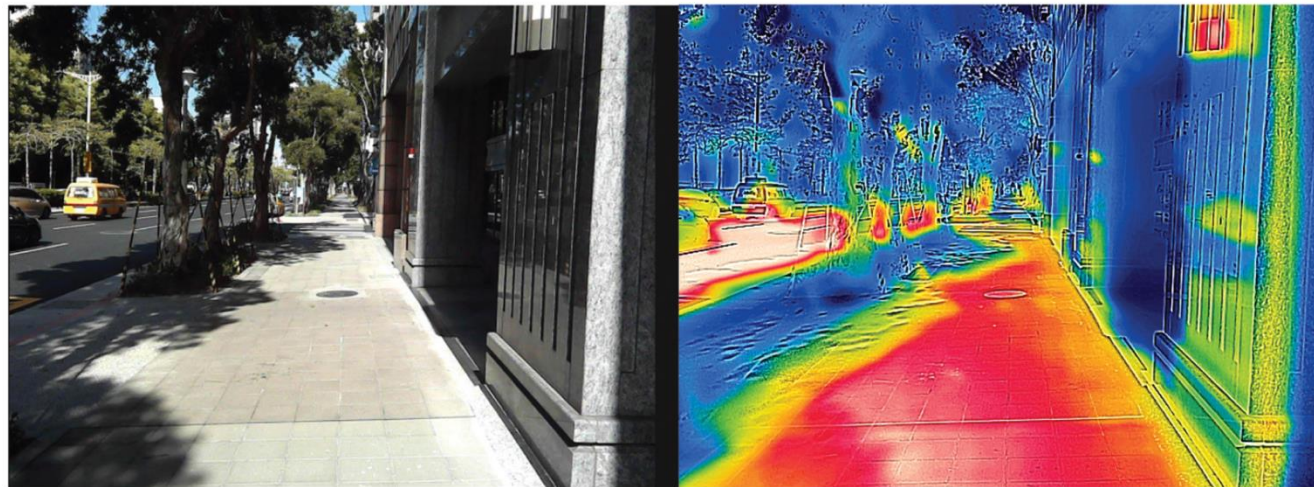
Infrared sensing

- Infrared sensors can be used to measure temperature
- They are used to detect cold and warm spots in buildings, and surface temperatures in various environments



Identifying and classifying thermal anomalies in Arctic houses using two infrared cameras and different metrics

<https://doi.org/10.1016/j.buildenv.2025.113101>

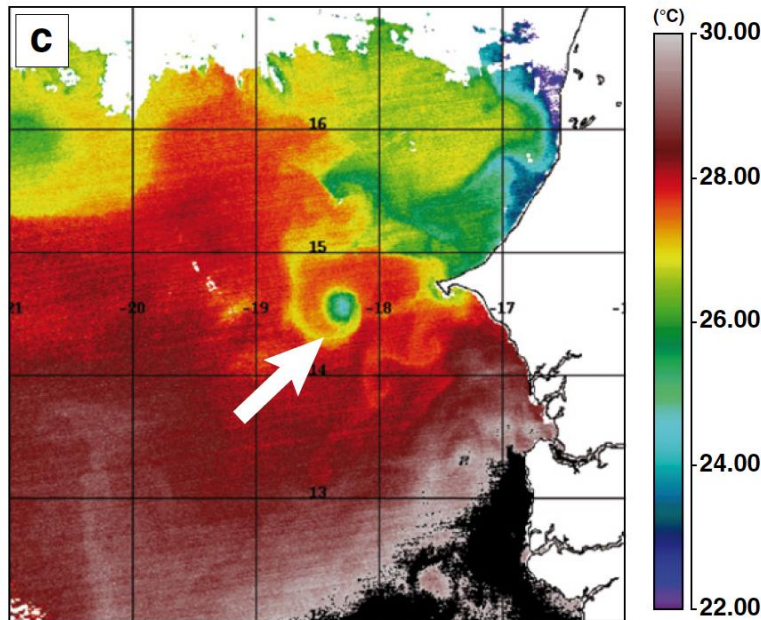


Influence of the thermal environment of urban sidewalks under green shading from a human scale <https://doi.org/10.1080/27669645.2024.2327253>

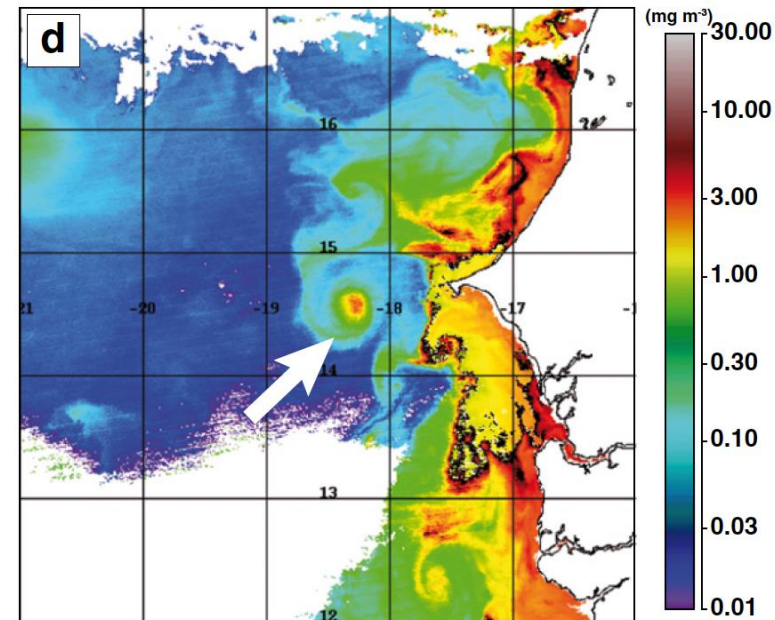
Sea surface temperature from infrared

- Sea surface temperature can provide evidence of warming
- It can be used to identify eddies, fronts and other features
- With other types of data, it can be used to understand scenarios
- For example, following a warm eddy (below)

SST (°C) MODIS 2011/11/07 14:35

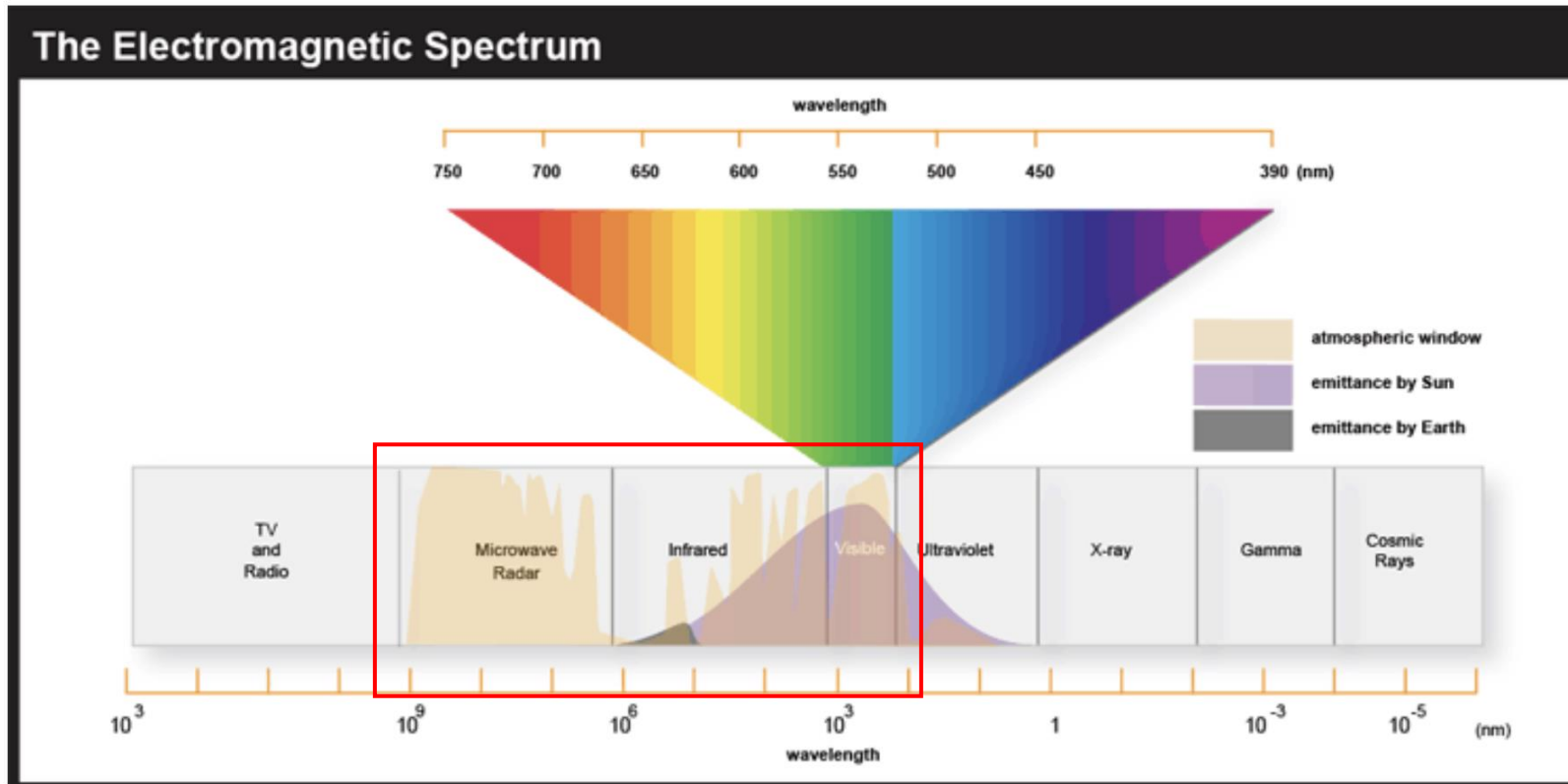


CHL (mg m⁻³) MODIS 2011/11/07 14:35



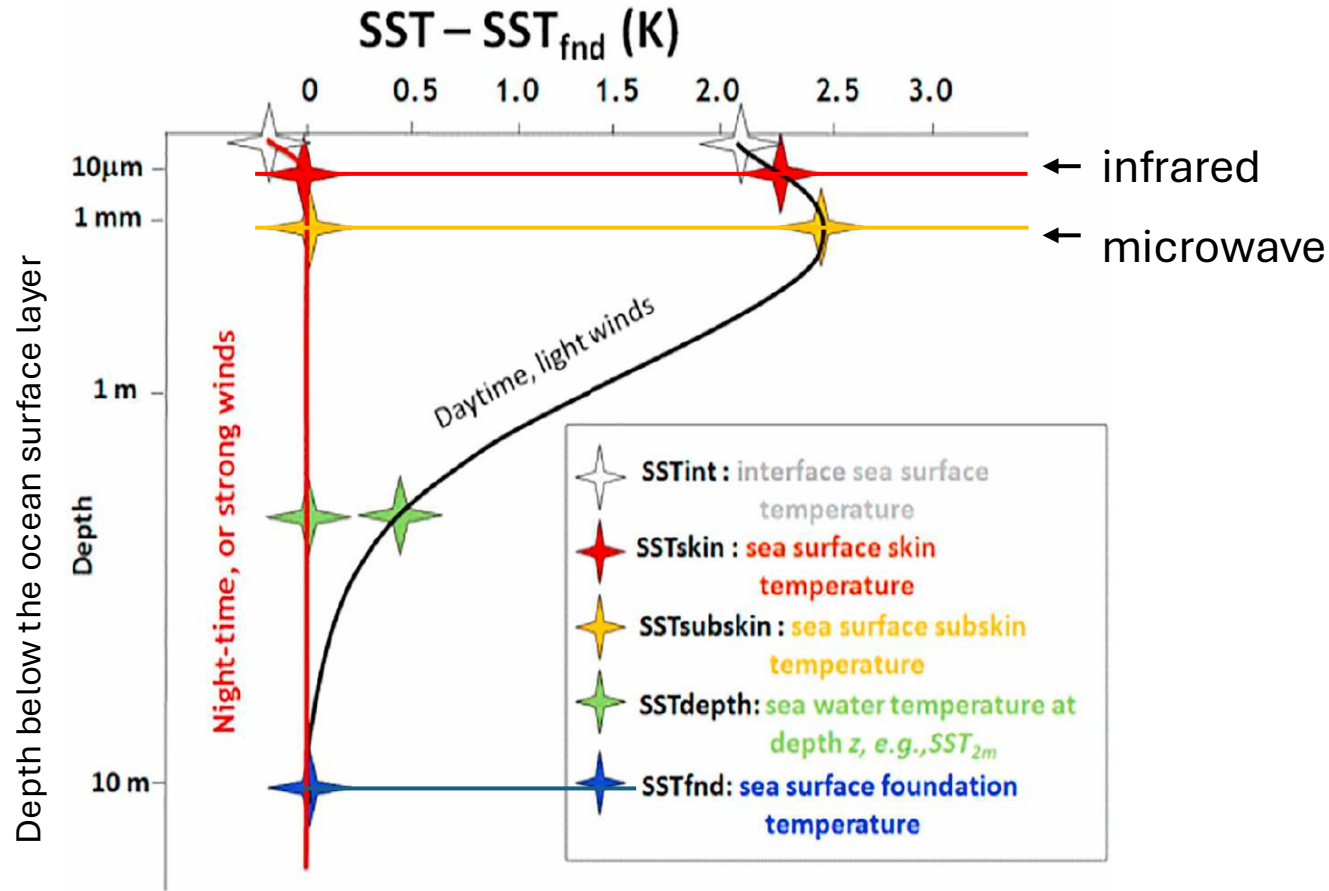
A small-scale oceanic eddy off the coast of West Africa studied by multi-sensor satellite and surface drifter data <http://dx.doi.org/10.1016/j.rse.2012.10.032>

Recall the electromagnetic spectrum



- remote sensing of the earth's surface uses "atmospheric windows" which have a high transmissivity -> transmit radiation without absorption

What is being measured?

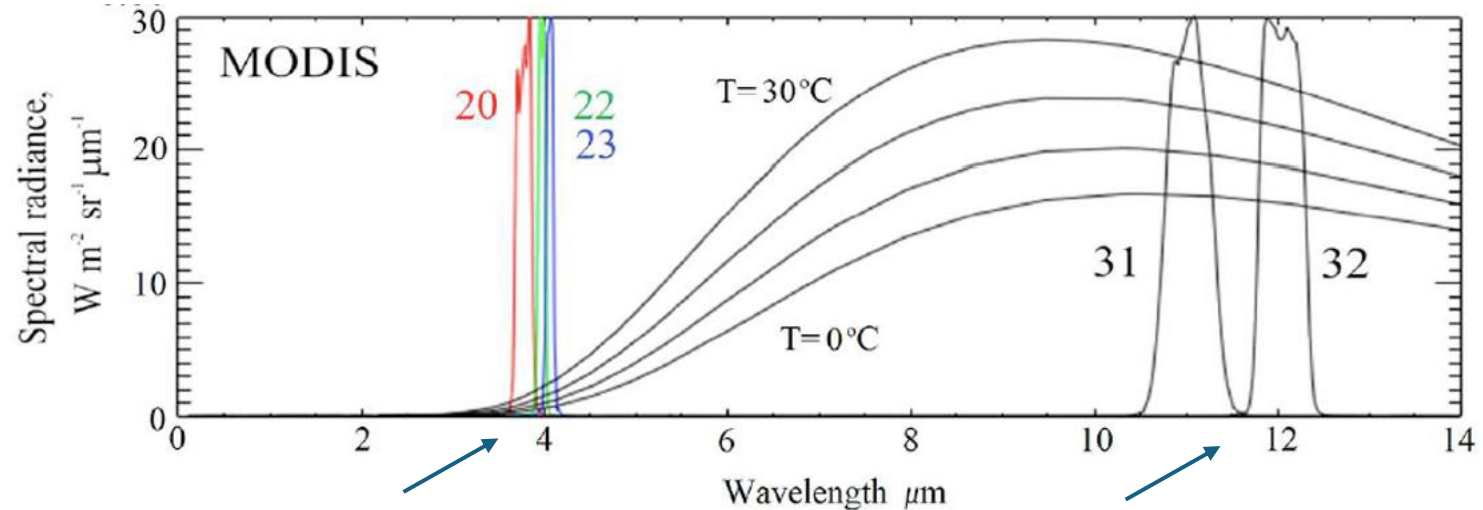


Minnett et al., 2019

- Recall the depth of the emitting layer is frequency dependent
- emission in the thermal infrared is from a “skin layer”
- emission in the microwave regime is from a slightly deeper “subskin layer”
- Both are closer to the surface than the “foundation temperature” – which is the depth that should not be influenced by daily heating/cooling

Wavelength selection for SST measurement

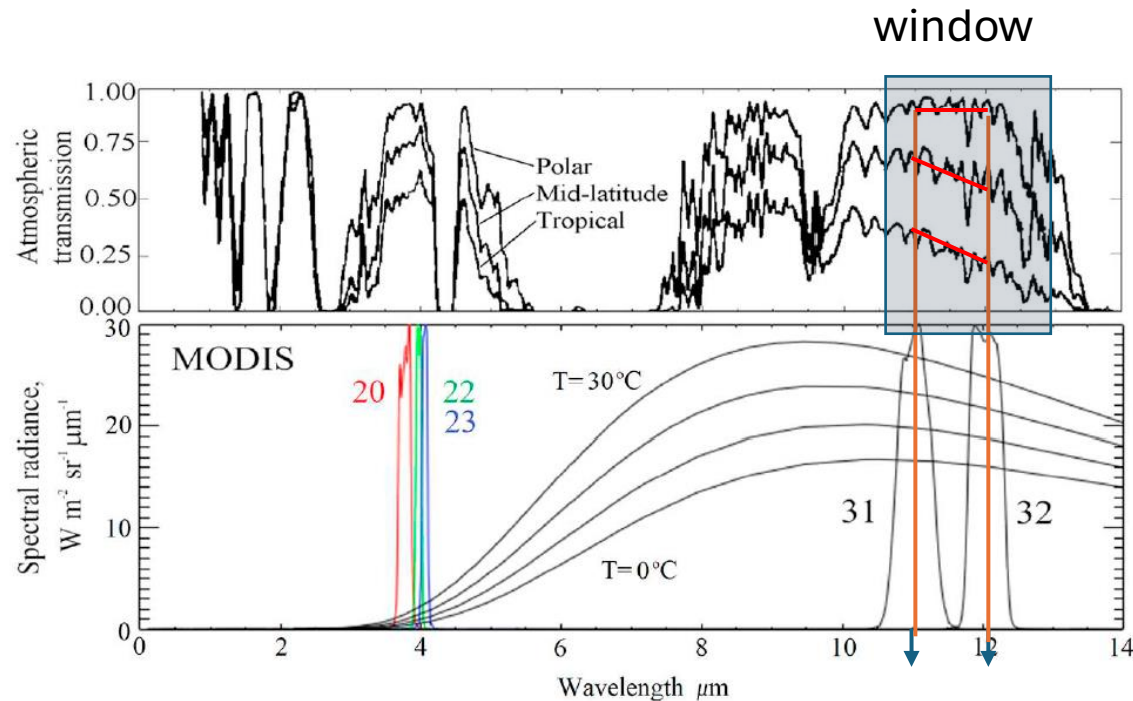
- wavelengths corresponding to thermal/infrared channels are used for SST



These channels partially pick up reflections so they are used only for nighttime SST

These channels can be used day and/or night

Atmospheric transmission and SST measurement



higher atmospheric transmission $\rightarrow$ less attenuation of surface signal

Minnett et al., 2019

- Differences in transmission at $11\mu\text{m}$ and $12\mu\text{m}$ tells us about atmosphere attenuation (usually due to water vapour)

Overview of the 'split window technique'

- consider an atmospheric window -> range of wavelengths for which transmissivity is high (atmospheric absorption is low)
- consider two measurements at two different wavelengths in this window
- absorption processes at these two wavelengths must be similar
- this allows linearization of Planck's Law (McMillin 1984), arrive at

$$T_s = T_1 + (T_1 - T_2)\gamma$$

- T_1, T_2 are the brightness temperatures at the two wavelengths (channels)
- T_s is the surface temperature, γ is a normalized atmospheric absorption coefficient

Inverting Planck's law to obtain brightness temperature

- Start with Planck's law

$$R(\lambda) = \frac{2hc^2}{\lambda^5} \left[\frac{1}{e^{hc/\lambda kT} - 1} \right]$$

- Rearrange

$$T = \frac{hc}{k\lambda} \left[\frac{1}{\ln \left(\frac{2hc^2}{\lambda^5 R(\lambda)} + 1 \right)} \right]$$

- Solve the above equation for T_1 and T_2 given λ_1 and λ_2 and $R(\lambda_1), R(\lambda_2)$
- Use in an equation of the form:

$$T_s = T_1 + (T_1 - T_2)\gamma$$

Linear multi-channel algorithm

- The basic linear algorithm is as follows

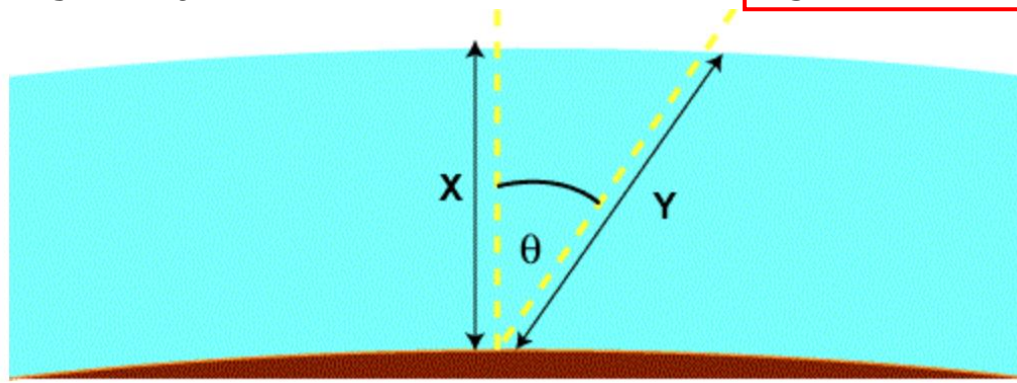
$$T_s = a_o + a_1 T_1 + a_2 (T_1 - T_2)$$

- a_o, a_1, a_2 are seasonally and/or regionally varying, determine using
 - match-ups (in-situ, within 1 hr and 10 km)
 - radiative transfer model
- a_2 is a function of transmissivity of the atmosphere
 - water vapour and temperature
- Two modifications were later made to improve accuracy
 - A correction for longer path lengths (satellite zenith angle)
 - A correction to account for local conditions for water vapour

Nonlinear SST algorithm

Satellite angle correction

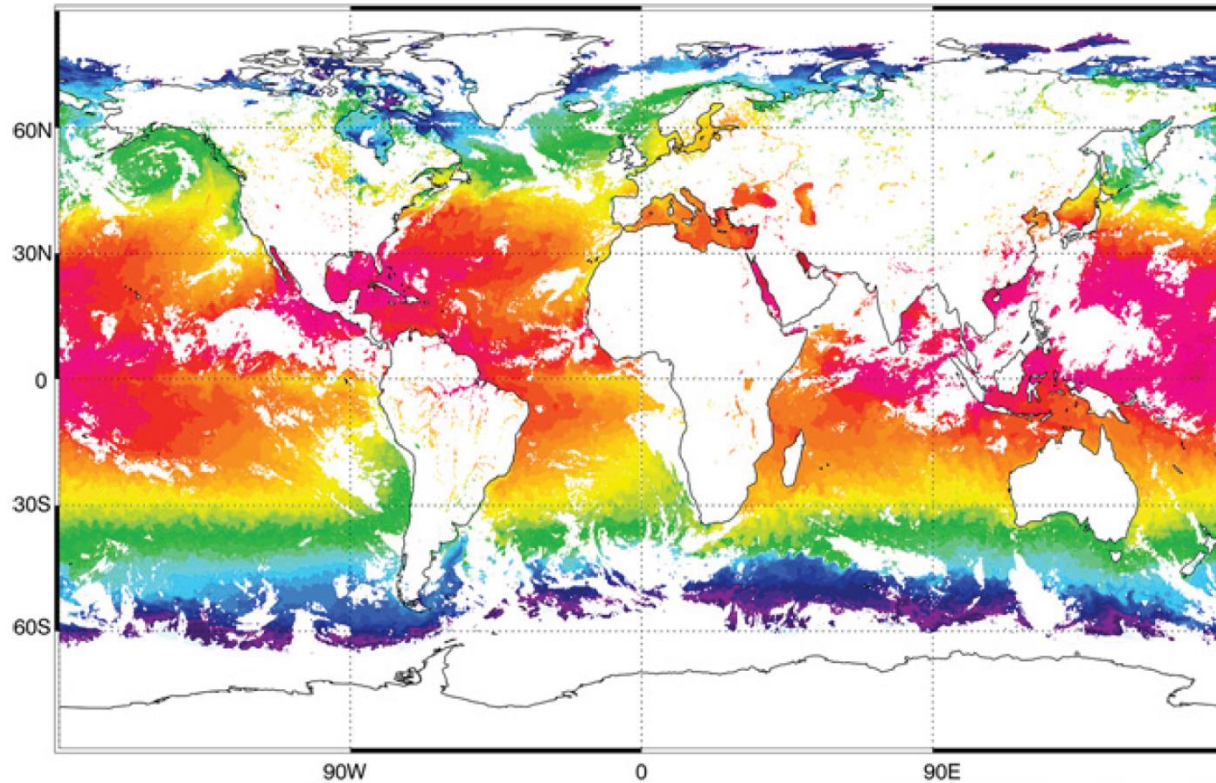
$$T_s = a_o + a_1 T_1 + a_2 (T_1 - T_2) + a_3 (\sec \theta - 1)$$



Correction for local conditions

$$T_s = a_o + a_1 T_1 + T_{sfc} (T_1 - T_2) + a_3 (\sec \theta - 1)$$

Examples of SST retrieval from infrared



- Fewer retrievals at high latitudes
- Patchy nature of the data
- Often multiple swaths are combined to fill in gaps -> this smears the temporal precision, but some sensors are on more than one satellite so have multiple passes/day

Sea surface temperature (SST)

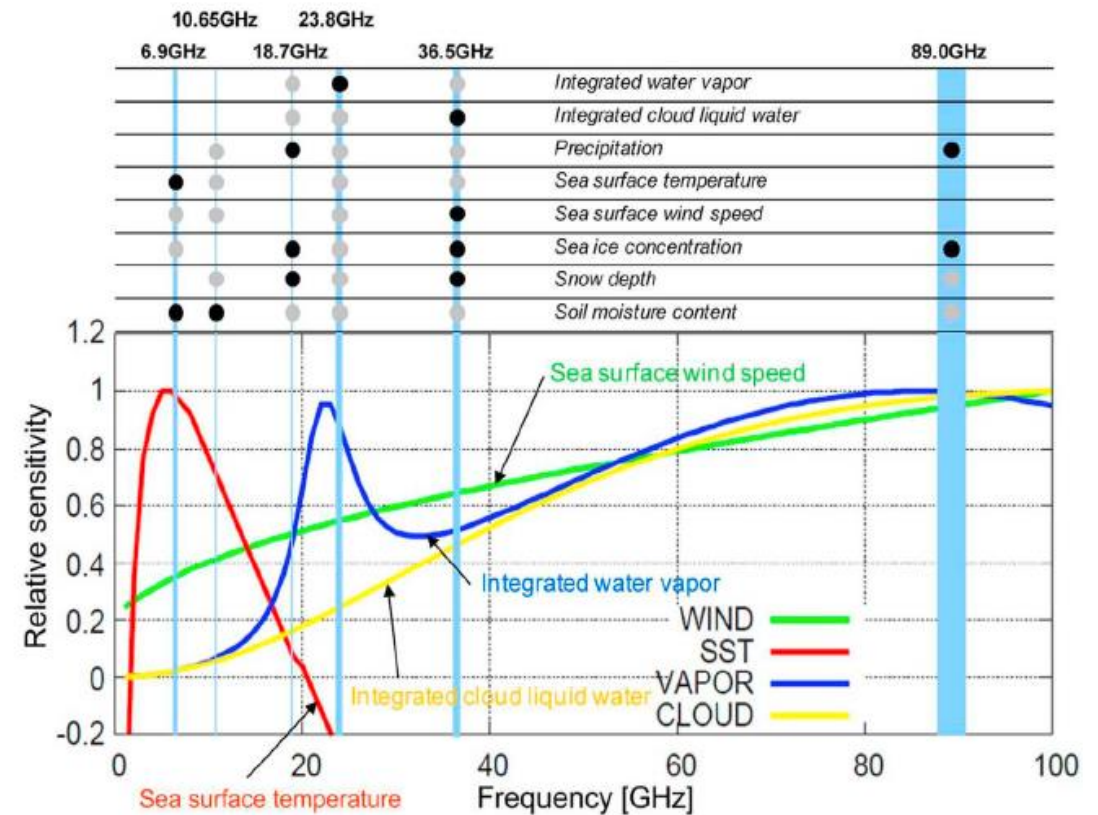
Microwaves

Passive microwave data

- passive microwave data provide near continuous coverage of the **earth's surface**
 - sunlight is not required
 - cloud cover does not block emission
 - data record extends over 40 years

Can be used to retrieve

- surface temperature
- windspeed
- salinity
- water vapor
- cloud liquid water
- sea ice concentration



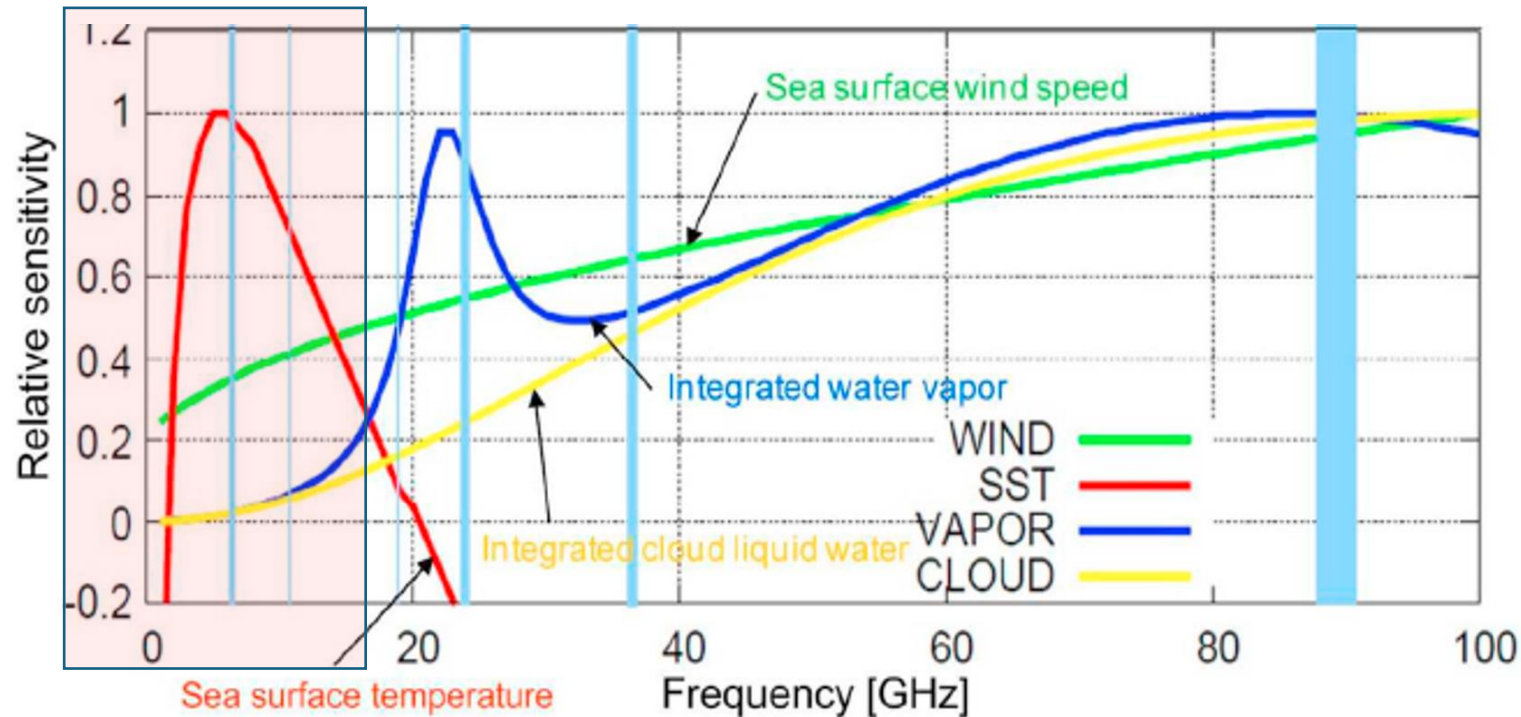
SST Retrieval from passive microwave data

Two common approaches

- statistical methods
- physically based approach (inversion of radiative transfer model)

SST from passive microwave: Statistical approach

- recall: there is significant overlap between sensitivity to SST and windspeed
- these two geophysical quantities are retrieved together in two stages



SST from passive microwave: Statistical approach

First stage retrieval

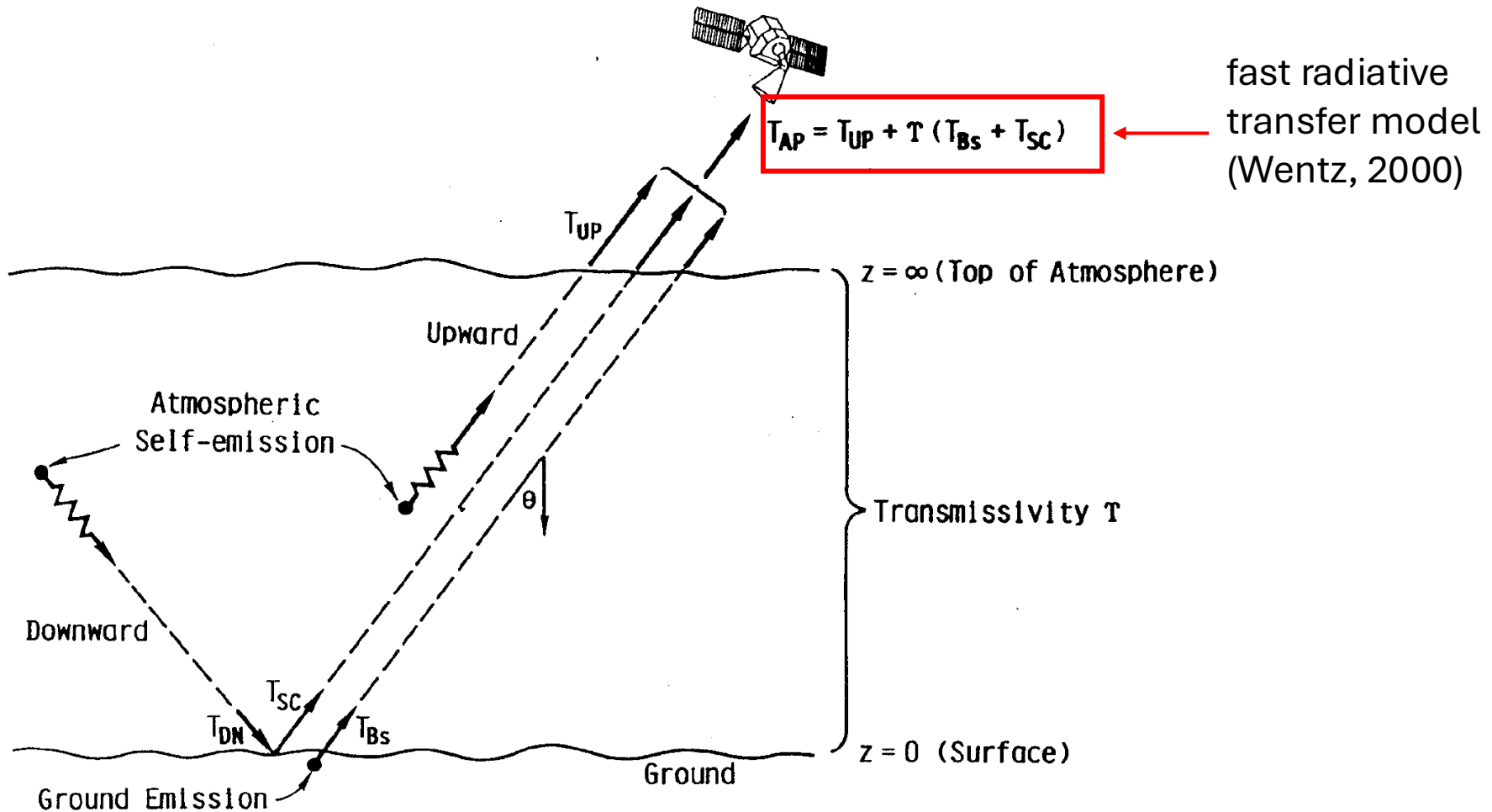
$$p_{1j} = a_{oj} + \sum_{i=1}^{Nchannels} a_{ij}t_i + b_{ij}t_i^2$$

- $j=1,2$ for SST or windspeed
- t_i is brightness temperature for channel i
- a_{ij}, b_{ij} are regression coefficients

Second stage retrieval

- additional “localized” regression across sets of windspeed and SST bins
- first stage is used to locate the observation in a bin
- latitudinal dependence and relative wind direction sometimes also considered

SST from passive microwave: physical model



Based on Ulaby et al. 1986

SST from passive microwave: physical model

$$\mathbf{y} = \boxed{H(\mathbf{x})} + \mathbf{e}$$

$\mathbf{y}$ vector of brightness temperatures

$\mathbf{x}$ state vector (windspeed, SST

$\mathbf{e}$ error vector

$H(\)$ radiative transfer model

- Objective: find the best estimate of the state given the observations
- Ill-posed inverse problem (Rodgers 2002)

$$\mathbf{x} = H^{-1}(\mathbf{y}) + \dots$$

- Optimal interpolation is used (Nielsen-Englyst, 2021)

SST from passive microwave: physical model

minimize a cost function of the form

$$J[\mathbf{x}] = \underbrace{(\mathbf{x} - \mathbf{x}_b)^T S_b^{-1} (\mathbf{x} - \mathbf{x}_b)}_{\text{background term}} + \underbrace{(\mathbf{y} - H(\mathbf{x}))^T S_{obs}^{-1} (\mathbf{y} - H(\mathbf{x}))}_{\text{observation term}}$$

$\mathbf{x}$ **optimal** state vector
 $\mathbf{x}_b$ **background** state vector

} windspeed, SST, cloud liquid water, water vapour...

$\mathbf{y}$ observation vector, (instrument channels, 6.9 GHz, 10.8 GHz)

S_b background error covariance matrix

S_{obs} observation error covariance matrix

$H(\mathbf{x})$ forward model (radiative transfer model)

SST from passive microwave: Uncertainty

Given a cost function of the form

$$J[\mathbf{x}] = (\mathbf{x} - \mathbf{x}_b)^T S_b^{-1} (\mathbf{x} - \mathbf{x}_b) + (\mathbf{y} - H(\mathbf{x}))^T S_{obs}^{-1} (\mathbf{y} - H(\mathbf{x}))$$

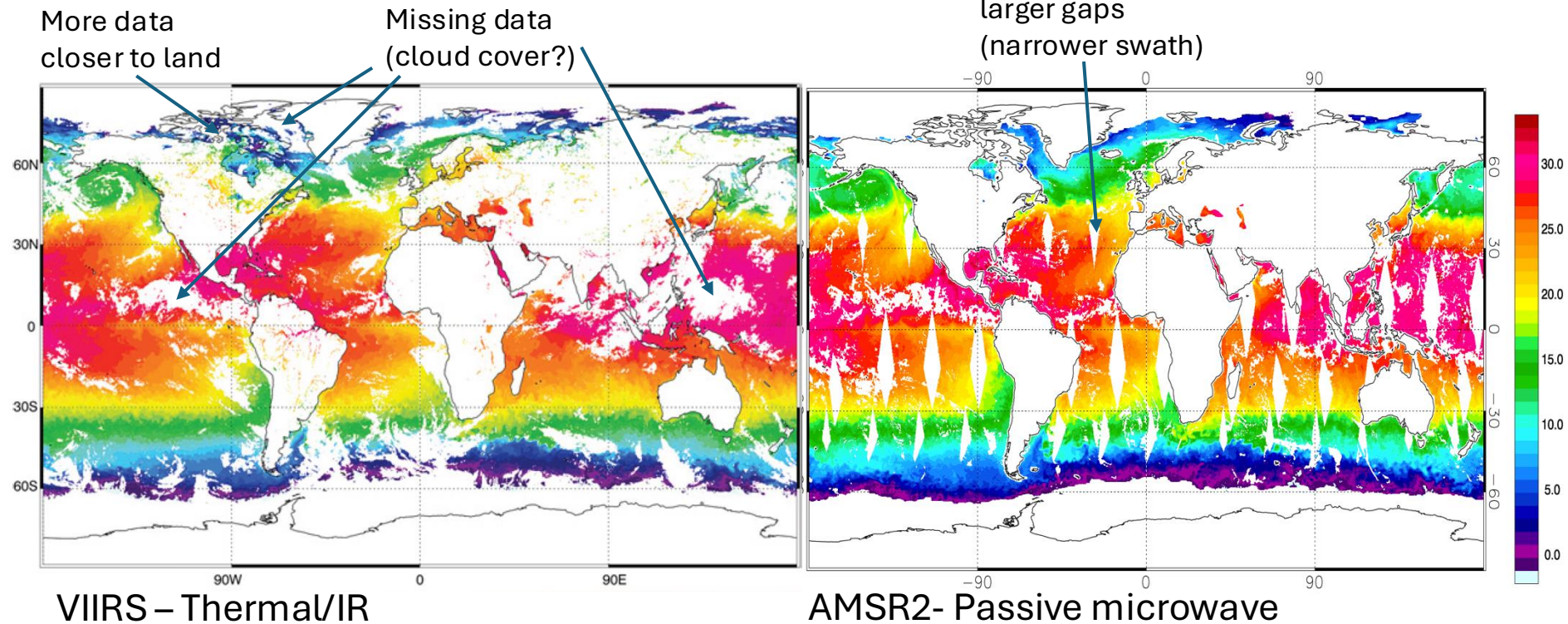
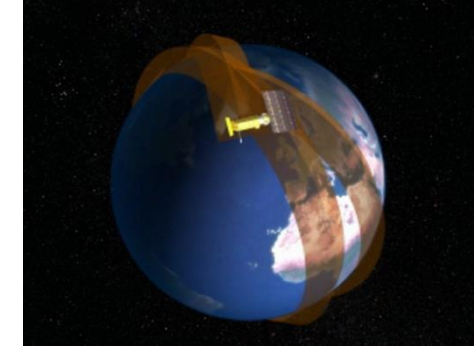
The error in the optimal state estimate can be expressed as

$$A = [S_b^{-1} + \mathbf{H}^T S_{obs}^{-1} \mathbf{H}]^{-1}$$

Sometimes used as a proxy for uncertainty

(can be found in some data files)

SST from thermal/infrared vs. passive microwave



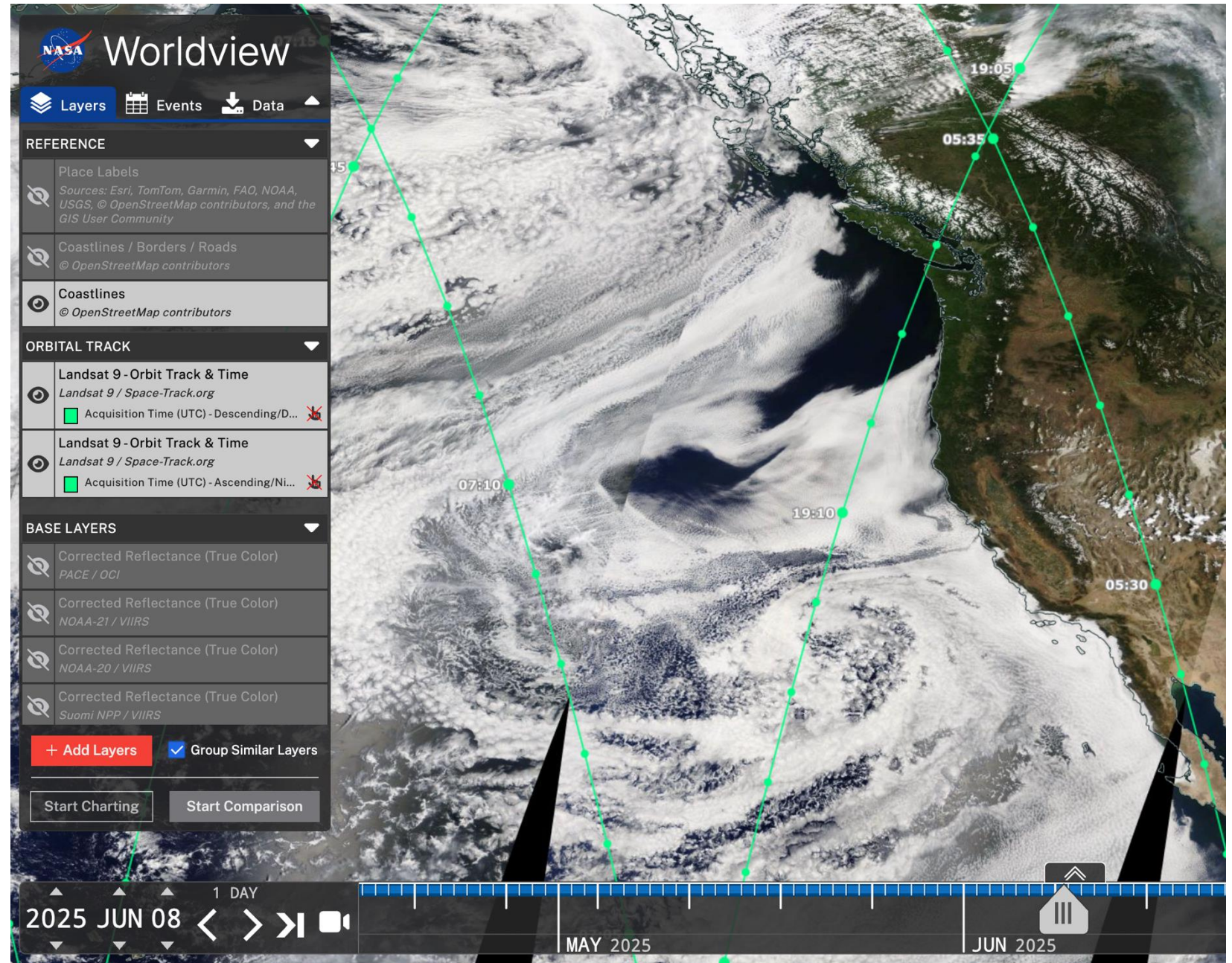
Sea surface temperature (SST) example

Study region:

You can start by looking at Worldview

You can turn on/off layers to look at different variables

For example, here I am looking at Landsat-9 tracks for June 2025 to find a clear image for our study region



Landsat data

Search Criteria **Data Sets** Additional Criteria Results

2. Select Your Data Set(s)

Check the boxes for the data set(s) you want to search. When done selecting data set(s), click the *Additional Criteria* or *Results* buttons below. Click the plus sign next to the category name to show a list of data sets.

☐ Use Data Set Prefilter [\(What's This?\)](#)

Data Set Search:

- ISERV
- Land Cover
- Landsat
 - Landsat Collection 2 Level-3 Science Products
 - Landsat C2 U.S. Analysis Ready Data (ARD)
 - Landsat Collection 2 Level-2
 - ☐ Landsat 8-9 OLI/TIRS C2 L2
 - ☐ Landsat 7 ETM+ C2 L2
 - ☐ Landsat 4-5 TM C2 L2
 - Landsat Collection 2 Level-1
 - ☒ Landsat 8-9 OLI/TIRS C2 L1
 - ☐ Landsat 7 ETM+ C2 L1
 - ☐ Landsat 4-5 TM C2 L1
 - ☐ Landsat 1-5 MSS C2 L1
 - Landsat C2 Atmospheric Auxiliary Data
 - Landsat Collection 2 DEM
 - Landsat Legacy

Started with level 1 data –
lower level, less processing

← → ↺ earthexplorer.usgs.gov

📧 Gmail 📺 YouTube 📍 Maps 🛡️ COVID-19 Map - J... 🌐 LAADS S&O - ima... 📺 New Music Friday:... 🏠 CPR 📺 Lecture 20 - Seco...

An official website of the United States government [Here's how you know](#)

USGS
science for a changing world

EarthExplorer Manage Criteria

Search Criteria **Data Sets** Additional Criteria Results

1. Enter Search Criteria

To narrow your search area: type in an address or place name, enter coordinates or click the map to define your search area (for advanced map tools, view the [help documentation](#)), and/or choose a date range.

Geocoder GeoJSON/KML/Shapefile Upload

Select a Geocoding Method
Feature (GNIS)

Search Limits: The search result limit is 100 records; select a Country, Feature Class, and/or Feature Type to reduce your chances of exceeding this limit.

US Features World Features

Feature Name
(use % as wildcard)

State
All

Feature Type
All

Show Clear

Polygon Circle Predefined Area

Degree/Minute/Second Decimal

1. Lat: 48° 54' 13" N, Lon: 126° 48' 15" W	✖
2. Lat: 49° 06' 20" N, Lon: 125° 06' 43" W	✖
3. Lat: 47° 56' 29" N, Lon: 125° 32' 26" W	✖
4. Lat: 48° 12' 46" N, Lon: 123° 06' 43" W	✖

Use Map Add Coordinate Clear Coordinates

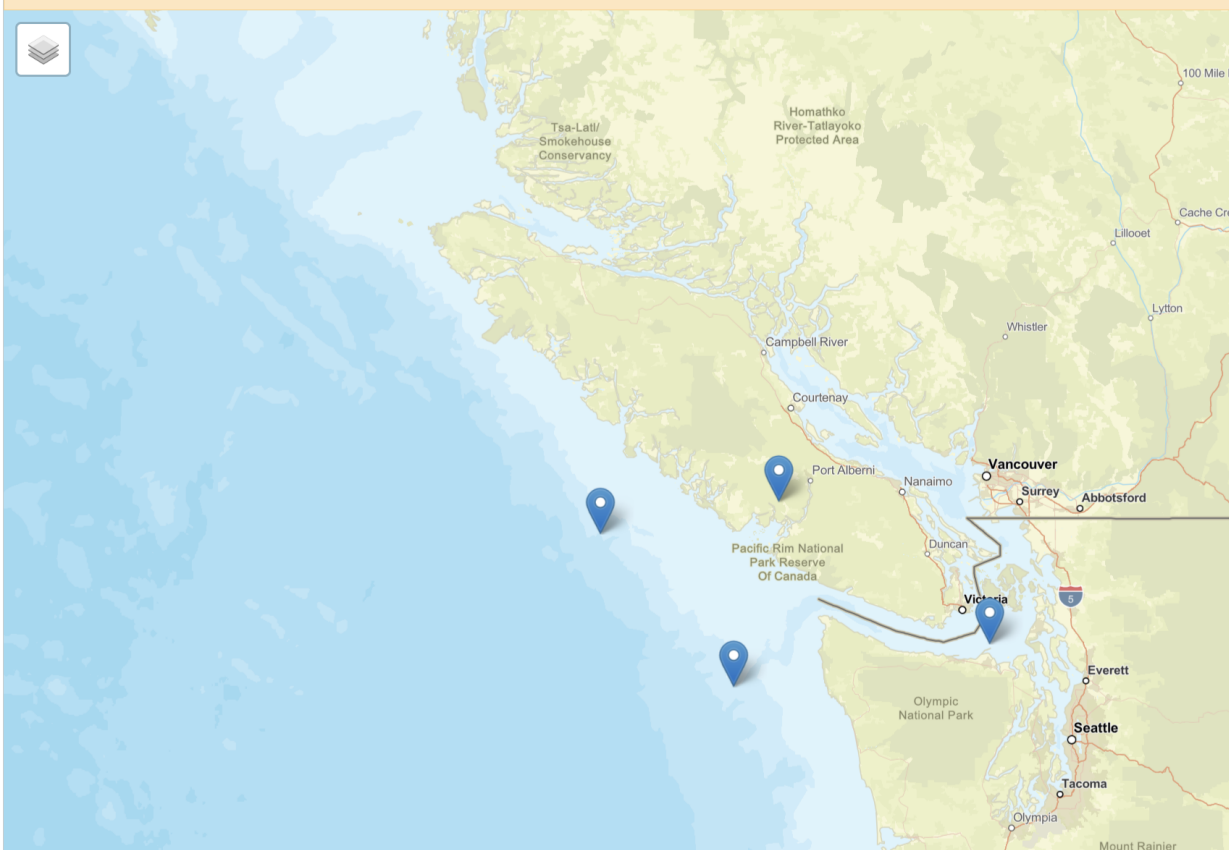
Date Range Cloud Cover Result Options

Search from: 06/01/2025 to: 07/30/2026

Search months: (all)

Data Sets » Additional Criteria » Results »

Search Criteria Summary (Show)



The provided maps are not for purchase or for download; they are to be used as a guide for reference and search purposes only; layers not owned or managed by USGS.

Sea surface temperature: Landsat data

To calculate sea – surface temperature

- Two bands:

Band 10 - Thermal Infrared (TIRS) 1	10.6–11.19	100 (resampled to 30)*
Band 11 - Thermal Infrared (TIRS) 2	11.50–12.51	100 (resampled to 30)*

- We could use a split window method (see tutorial)

$$T_s = a_o + a_1 T_1 + T_{sfc}(T_1 - T_2) + a_3(\sec\theta - 1)$$

- But Band 11 is degraded due to stray light
- Use a single channel approach
- Rearrange Planck's law to solve for brightness temperature

$$R(\lambda) = \frac{2hc^2}{\lambda^5} \left[\frac{1}{e^{hc/\lambda kT} - 1} \right]$$

- Need to adjust for emissivity and atmospheric contamination

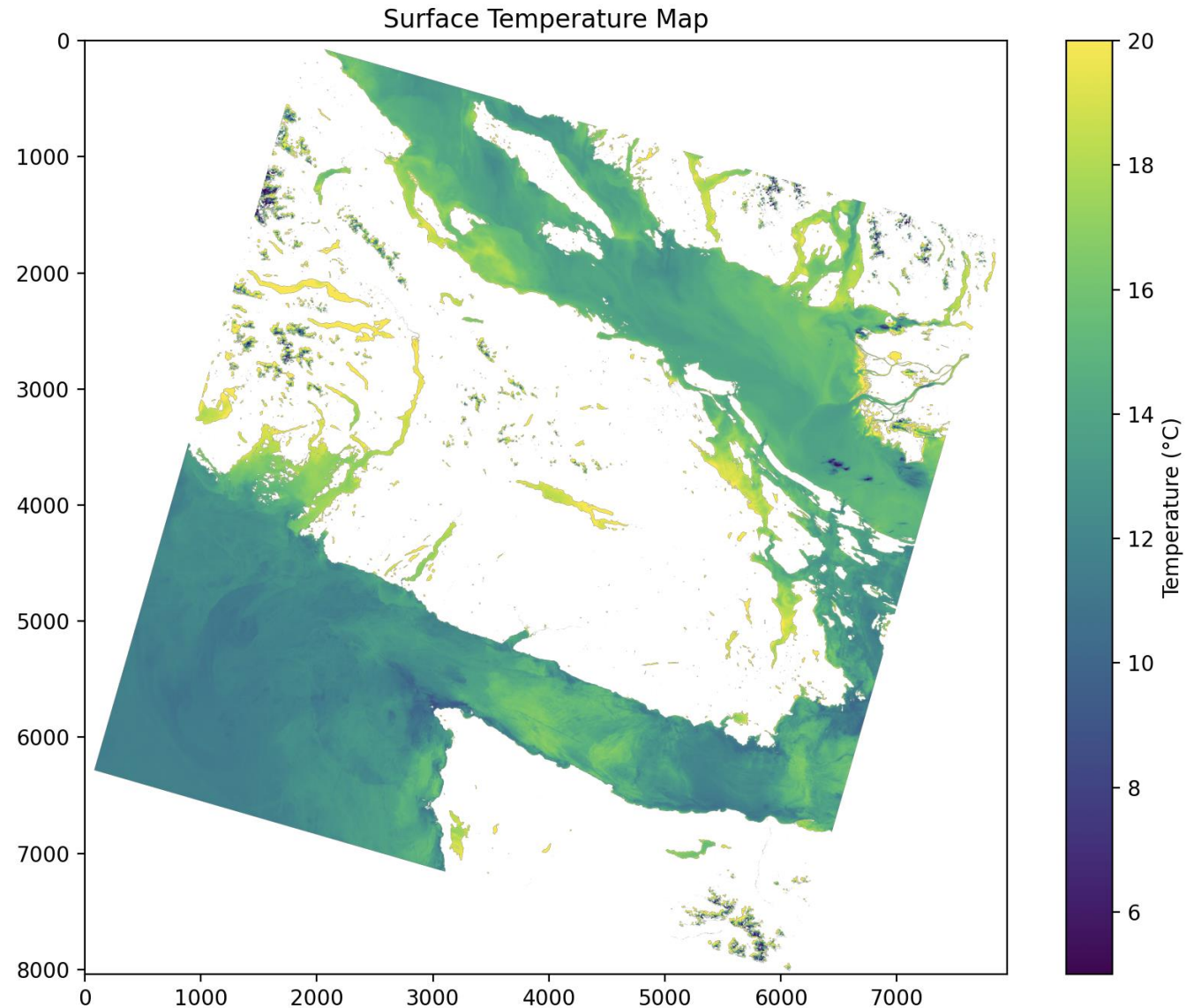
Landsat data – quality flags

- The file is: LC09_L2SP_048026_20250421_20250422_02_T1_QA_PIXEL.TIF
- This file has a 16 bit binary number for each pixel
- The bits are either on (1) or off (0) depending on if a condition is flagged
- Conditions are things like, cloud, land, snow etc.

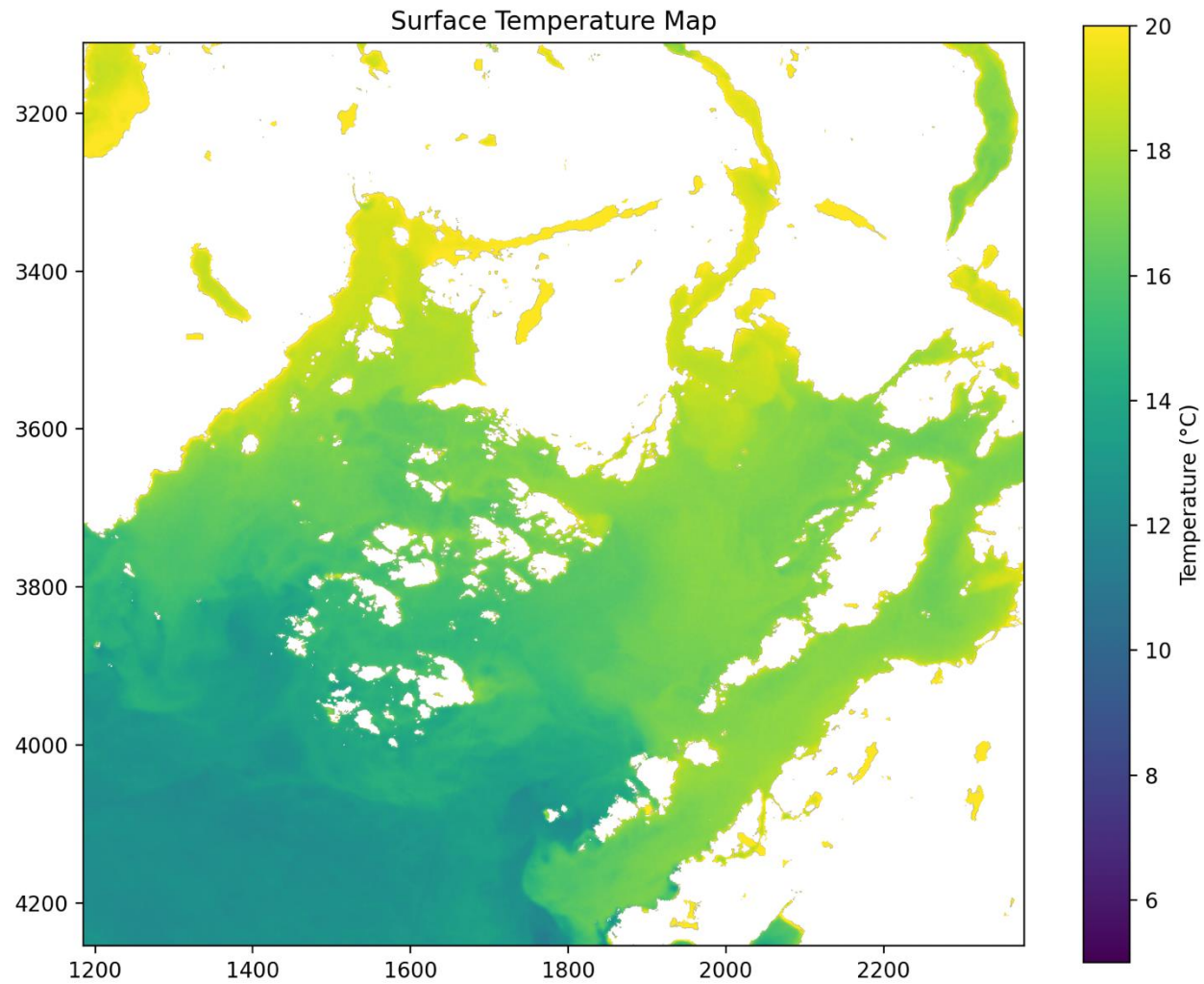
Sea surface temperature: Single channel

- Read Band 10 data
- Calculate radiance using scaling coefficients in the MTL.txt file
 $\text{RADIANCE_MULT_BAND_10} = 3.8000\text{E-04}$
 $\text{RADIANCE_ADD_BAND_10} = 0.10000$
- Convert radiance to brightness temperature
-> invert Planck's Law
- Divide brightness temperature by emissivity
(very high, 0.99 for water)

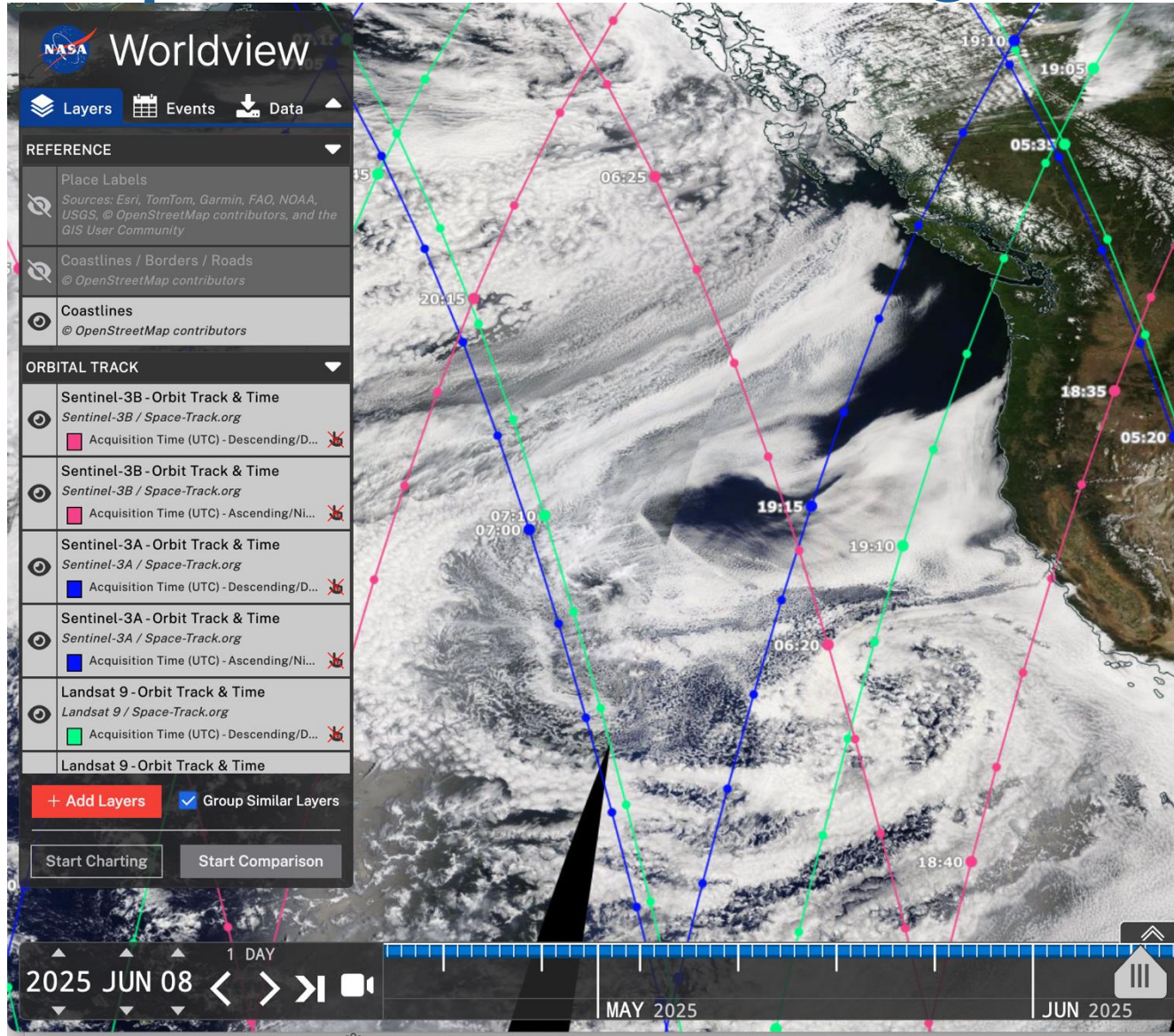
* Atmospheric correction – omitted here



Zoom into a coastal region:



Comparison with Sentinel 3



The screenshot shows the Copernicus Browser interface. The top navigation bar includes the Copernicus logo, language (EN), user (Andrea Scott), and a search button. Below the navigation bar, there are tabs for 'SH DASHBOARD' and 'WORKSPACE'. The main content area displays a list of data layers. The 'SENTINEL-3' layer is selected, and its sub-layers are listed: OLCI, SRAL, SLSTR, Level-1 RBT, Level-2 AOD, Level-2 FRP, Level-2 LST, Level-2 WST (selected), SYNERGY, Demo Products, and Auxiliary Data File. An arrow points from the text 'Level 2 WST: Water surface temperature' to the 'Level-2 WST' layer. Below the layers list, there is a 'TIME RANGE' section with 'From' and 'Until' date pickers. The 'From' date is 2025-06-08 and the 'Until' date is 2025-06-18.

Level 2 WST:
Water surface
temperature

<https://browser.dataspace.copernicus.eu/>

Sentinel-3 SLSTR (radiometer)

Band	Central Wavelength (nm)	Bandwidth (nm)	Function	Comments		Resolution (m)
S1	554.27	19.26	>Cloud screening, vegetation monitoring, aerosol	VNIR	Solar Reflectance Bands	500
S2	659.47	19.25	NDVI, vegetation monitoring, aerosol			
S3	868.00	20.60	NDVI, cloud flagging,Pixel co-registration			
S4	1374.80	20.80	Cirrus detection over land	SWIR		
S5	1613.40	60.68	loud clearing, ice, snow,vegetation monitoring			
S6	2255.70	50.15	Vegetation state and cloud clearing			
S7	3742.00	398.00	SST, LST, Active fire	Thermal IR Ambient bands (200 K -320 K)		1000
S8	10854.00	776.00	SST, LST, Active fire			
S9	12022.50	905.00	SST, LST			
F1	3742.00	398.00	Active fire	Thermal IR fire emission bands		
F2	10854.00	776.00	Active fire			

Main sensors on Sentinel-3

- SLSTR – radiometer (surface temperate)
- OCLI – spectrometer (ocean color)
- SRAL – altimeter (topography)

3.7 μm
11 μm
12 μm

3 infrared channels for surface temperature -> different methods

Different algorithms available for Sentinel 3

Dual-channel

- let T_{11} and T_{12} be the temperatures at $11\mu m$ and $12\mu m$

$$SST = ao + a_1 * T_{11} + a_2(T_{11} - T_{12}) * Tsfc$$

First guess of SST
atmospheric attenuation

There are also:

- Three channel
- Night-time
- Dual view!

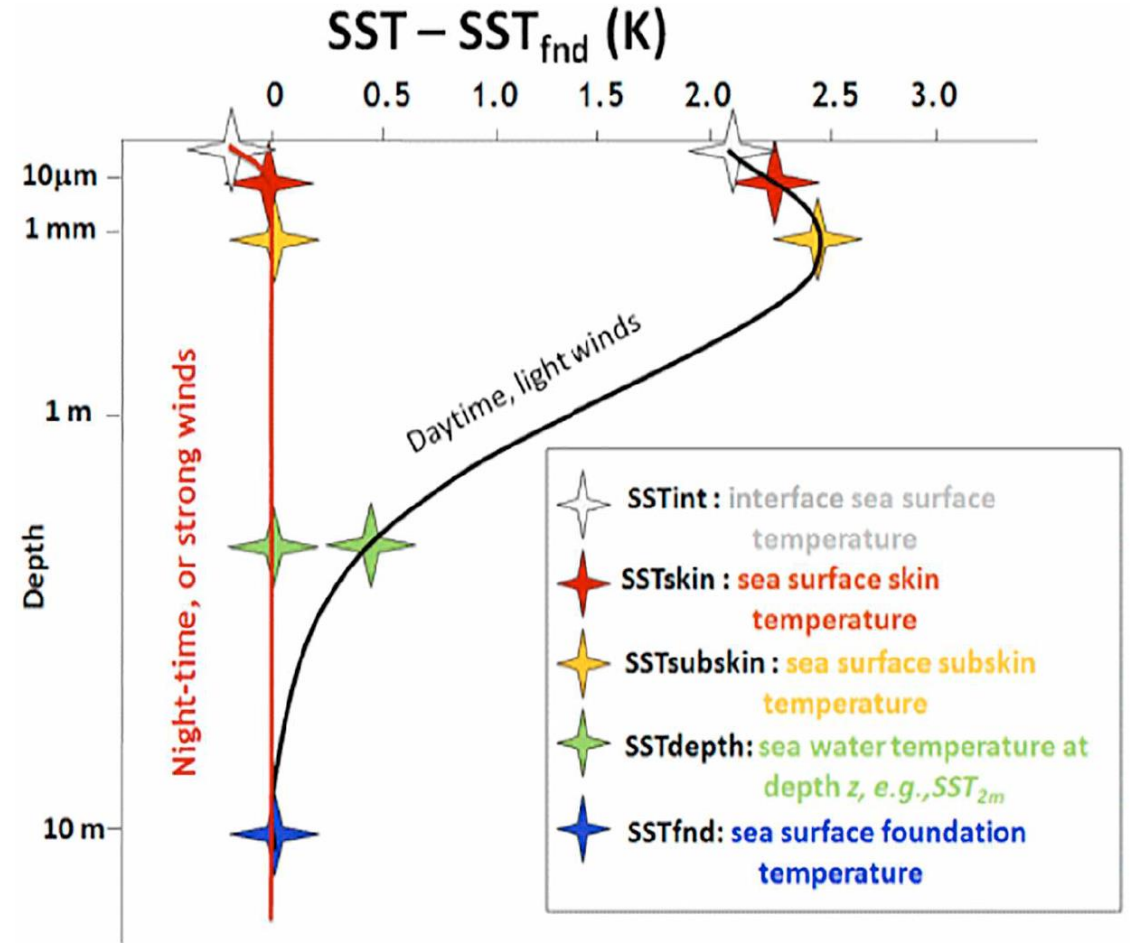
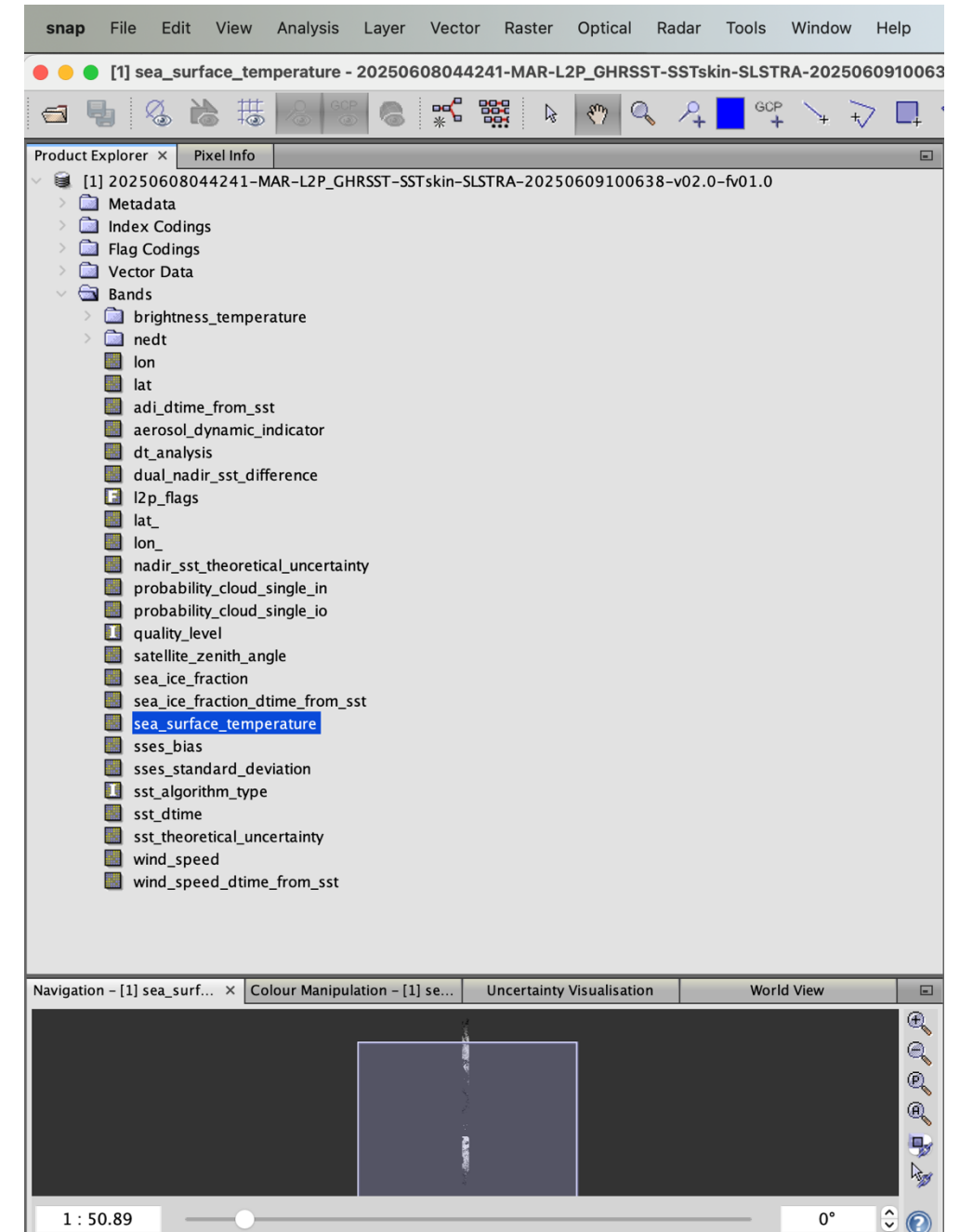


Fig. 2. Near-surface oceanic temperature gradients. From Minnett and Kaiser-Weiss (2012).

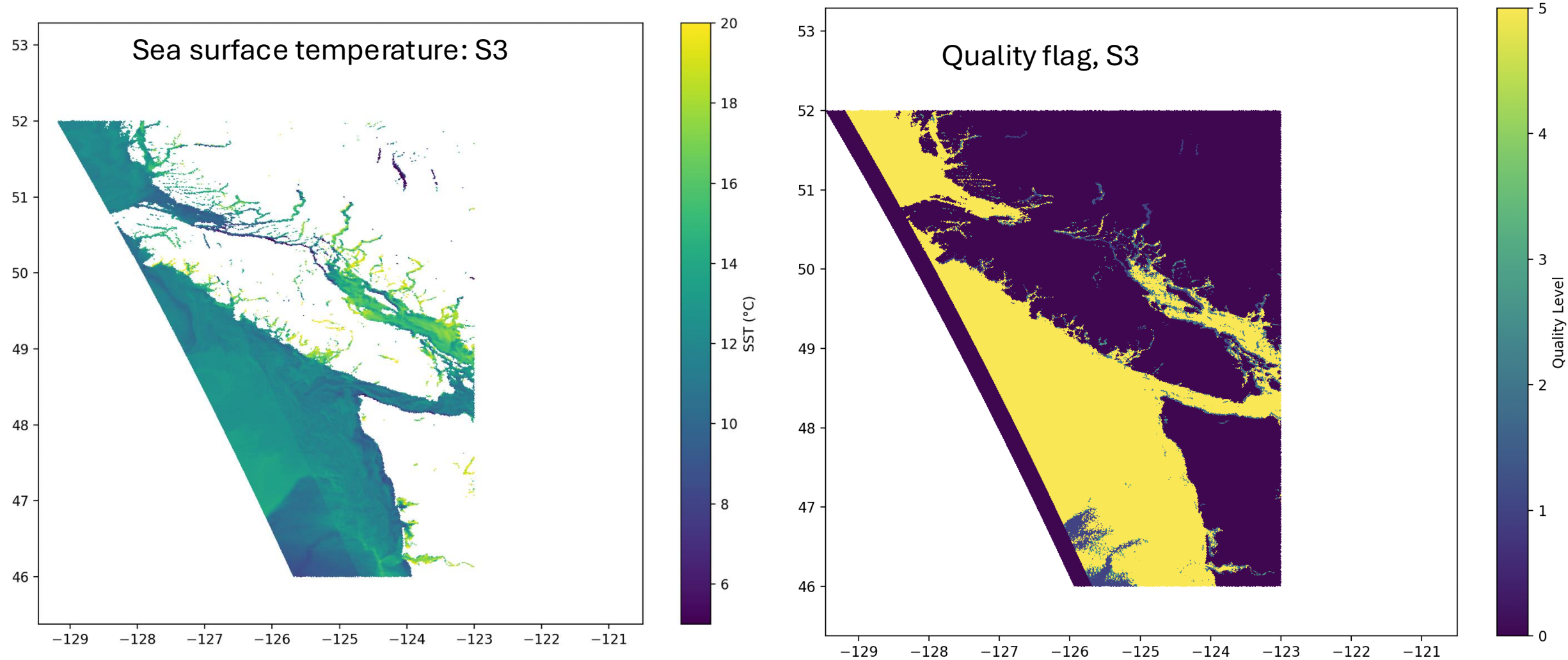
Sentinel 3: data file

There are several important variables in the netcdf file

- Sea surface temperature
- SST algorithm type
- Quality level
- Wind speed (from ERA5)
- Satellite zenith angle
- Aerosol dynamic indicator



Sea surface temperature from Sentinel 3



Make a note of what you see in each image

Comparing Sentinel 3 and Landsat SST

For Landsat the time is in the *MTL*.txt file

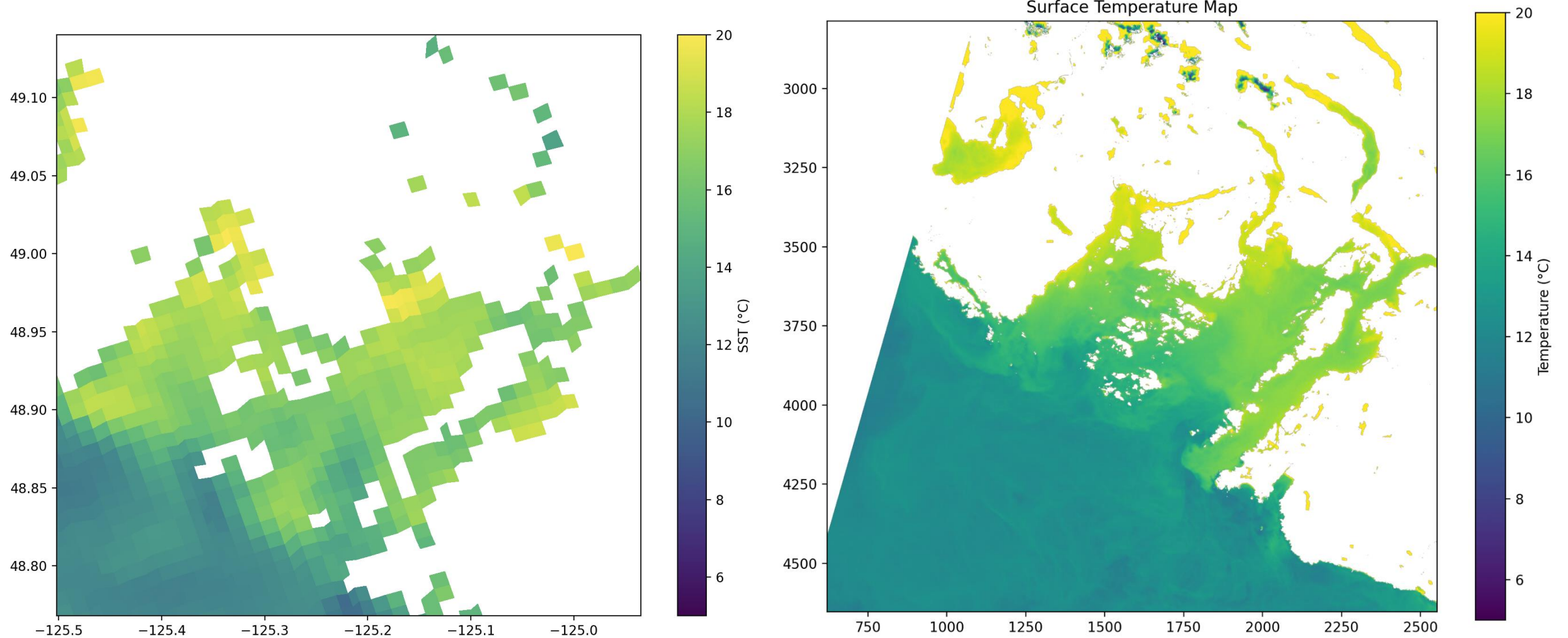
```
DATE_ACQUIRED = 2025-06-08  
SCENE_CENTER_TIME = "19:07:04.9109830Z"
```

For Sentinel 3 it is in the file name

20250608044241-MAR-L2P_GHRSST-SSTskin-SLSTRA-
20250609100638-v02.0-fv01.0.nc

2026-06-08 at 04:42:41

Sea surface temperature from Sentinel 3 vs Landsat



What are some differences between these?

Sea surface temperature (SST)

Data products

Recall the levels of satellite products

Level 0	Reconstructed, unprocessed instrument and payload data at full resolution (eg. sensor counts)
Level 1	Reconstructed, unprocessed instrument data at full resolution, time-referenced, and annotated with ancillary information, including radiometric and geometric calibration coefficients and georeferencing parameters, Level 1B are Level 1A data that have been processed to sensor units (eg. brightness temperature)
Level 2	Derived geophysical variables at the same resolution and location as L1 source data (eg. sea ice concentration)
Level 3	Variables mapped on uniform space-time grid scales, usually with some completeness and consistency (eg., gridded sea ice concentration)
Level 4	Model output or results from analyses of lower-level data (e.g., variables derived from multiple measurements, such as an ice velocity product that uses sea ice concentration).

paraphrased from:

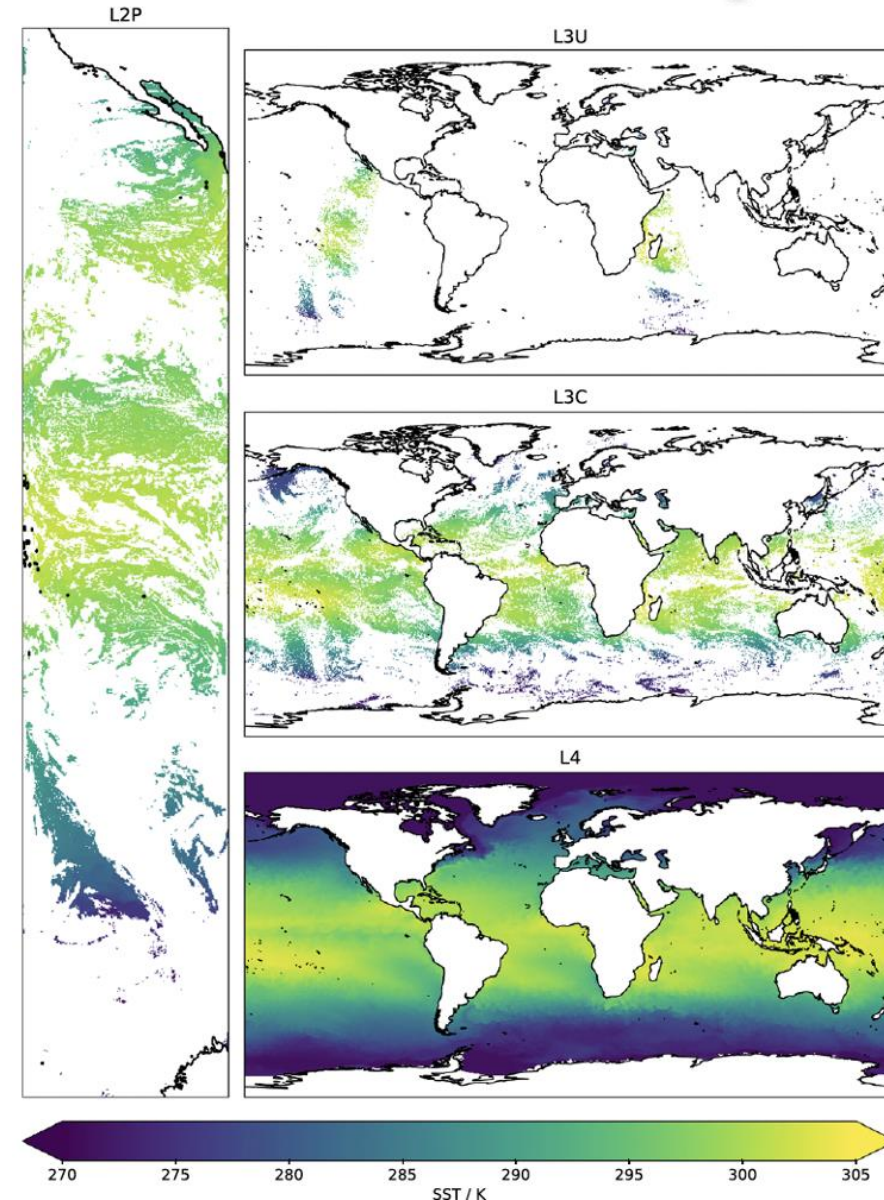
<https://earthdata.nasa.gov/collaborate/open-data-services-and-software/data-information-policy/data-levels>

Processing levels

The L2P SST product is derived at the native sensor resolution (~0.75 km at nadir, ~1.5 km at swath edge) using NOAA's Advanced Clear-Sky Processor for Ocean (ACSPo) system, and reported in 10-minute granules in netCDF4 format

SSTs are derived from Brightness Temperatures (BTs) using the Non-Linear SST (NLSST) algorithms (Petrenko et al., 2014)

From:
https://podaac.jpl.nasa.gov/dataset/VIIRS_NPP-OSPO-L2P-v2.3

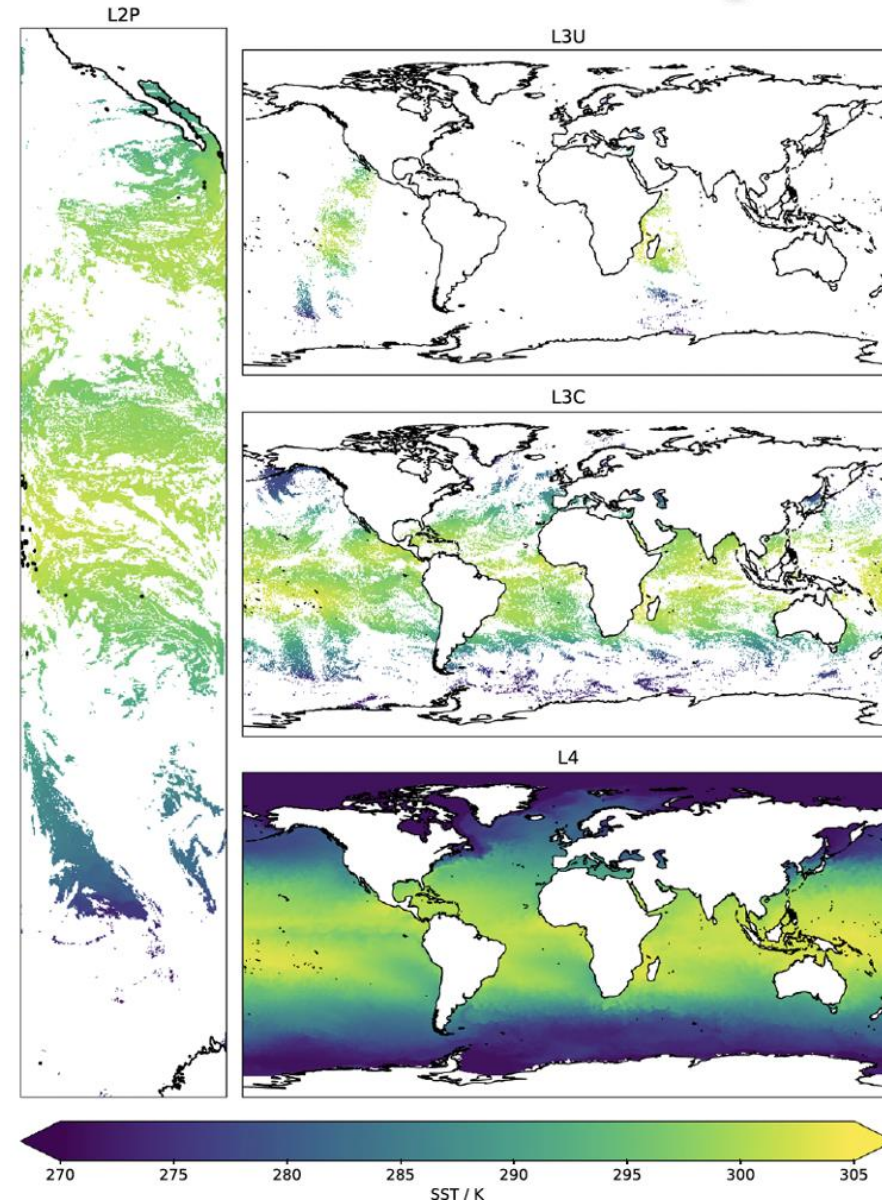


Processing levels

L3U – daily, gridded L2P

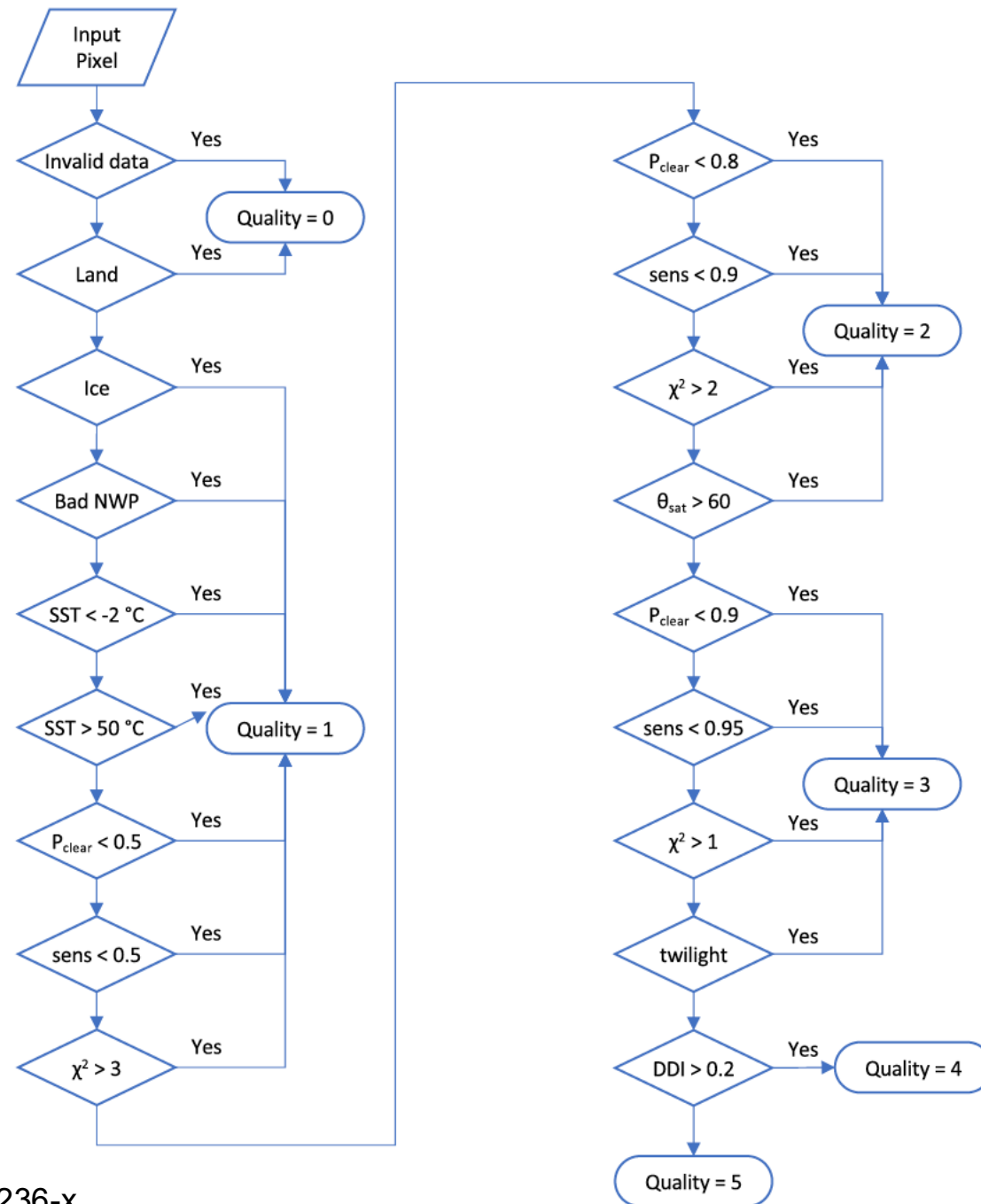
L3S – collocated data from different satellite missions (e.g. AVHRR and VIIRS)

L4 – daily, surface temperature analysis. Combines multi-satellite retrievals of sea surface temperature into a single analysis of SST. Used as a lower boundary condition in Numerical Weather Prediction (NWP) models.



Merchant, 2019

SST: Quality assessment



From Merchant et al.,
<https://doi.org/10.1038/s41597-019-0236-x>

On your own

- Compare glider temperature with satellite SST:
<https://cproof.uvic.ca/>
- Investigate the quality flags in the products
- Compile a multi-sensor dataset that captures diurnal variability
- Compare products from different processing levels
- Use SST or LST (land surface temperature) to investigate a scenario of interest